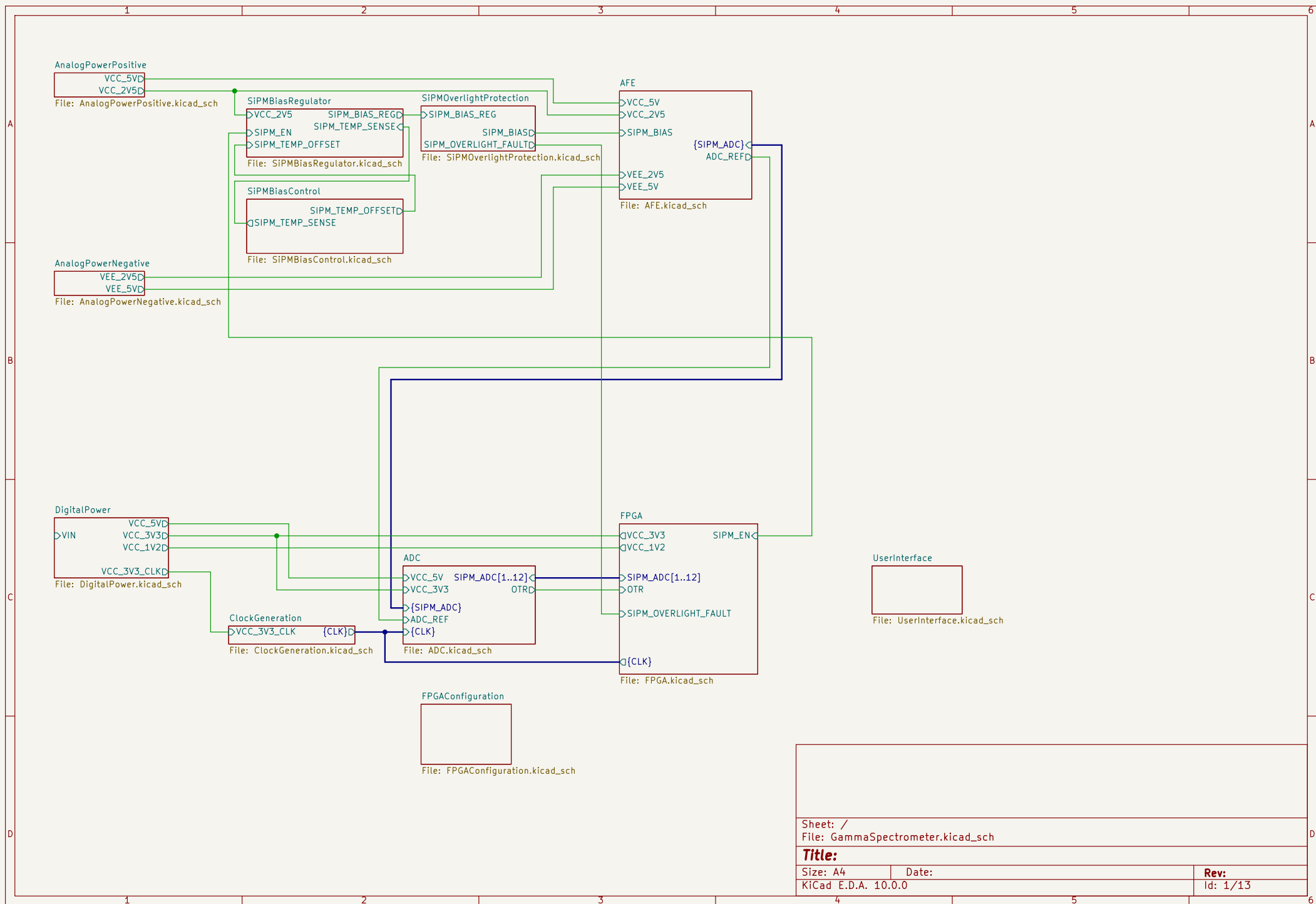
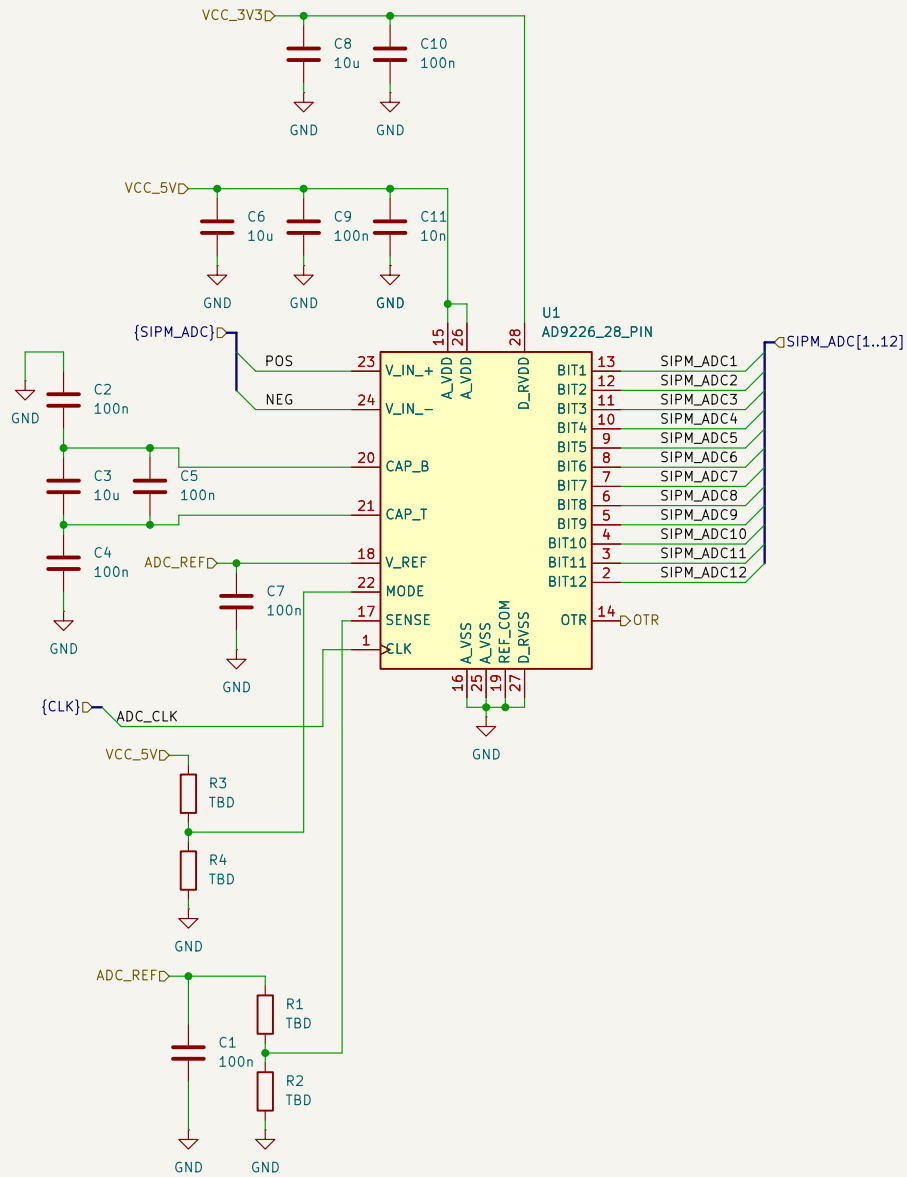






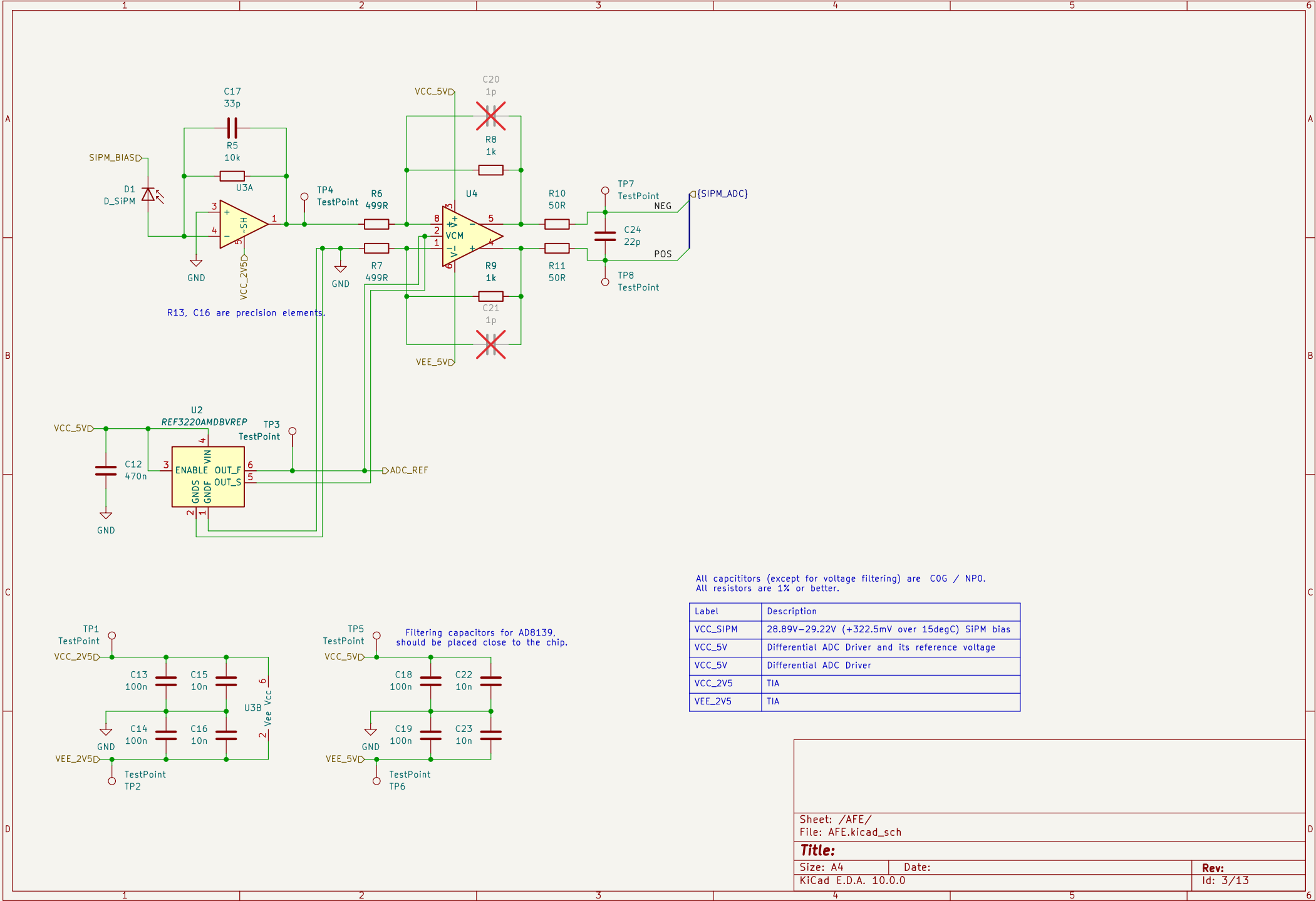
Sheet: /ADC/
File: ADC.kicad_sch

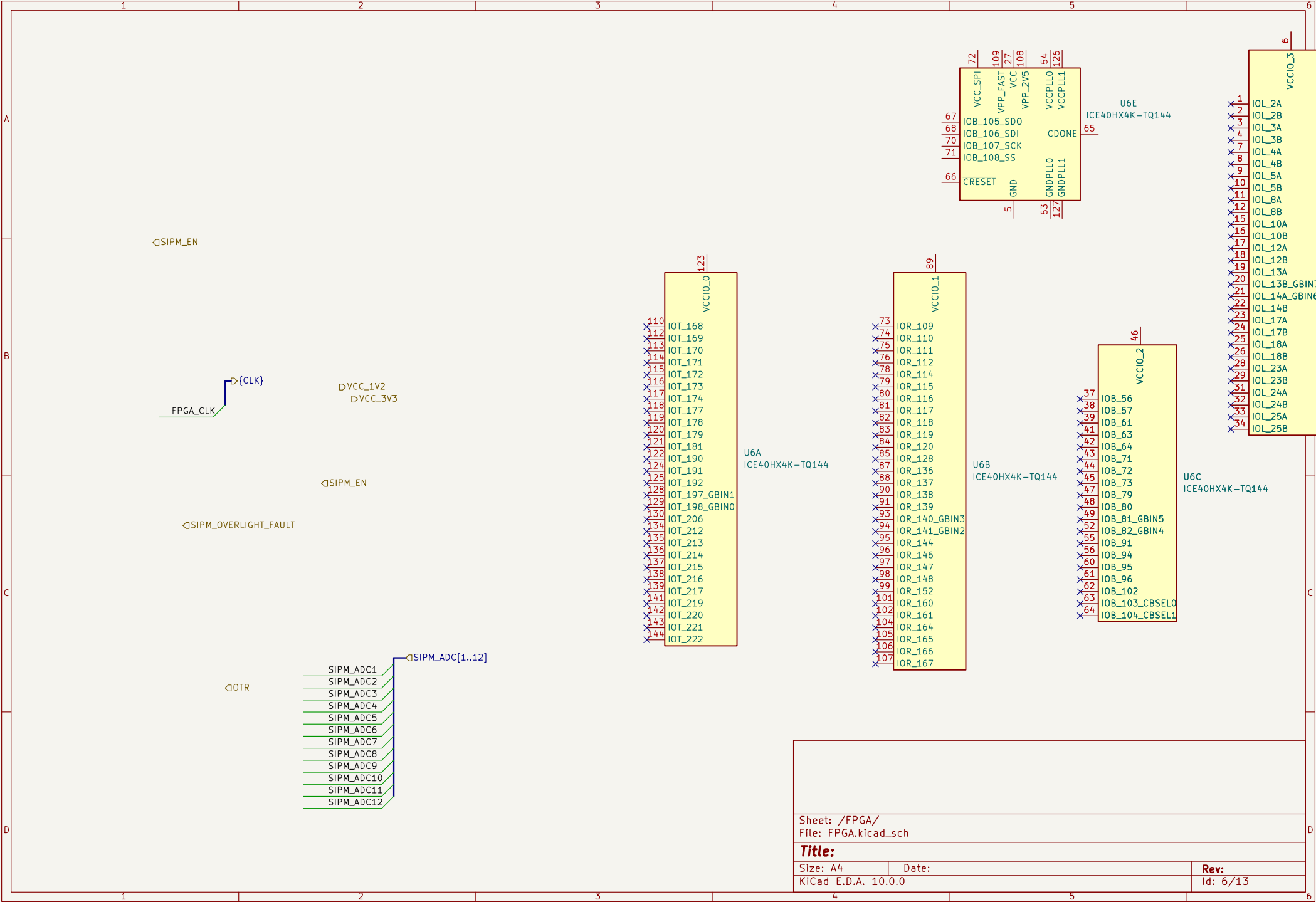
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Size: A4
KiCad E.D.A. 10.0.0

Date:

Rev:
Id: 2/13





Sheet: /FPGA/
File: FPGA.kicad_sch

Title:

Size: A4

Date:

KiCad E.D.A. 10.0.0

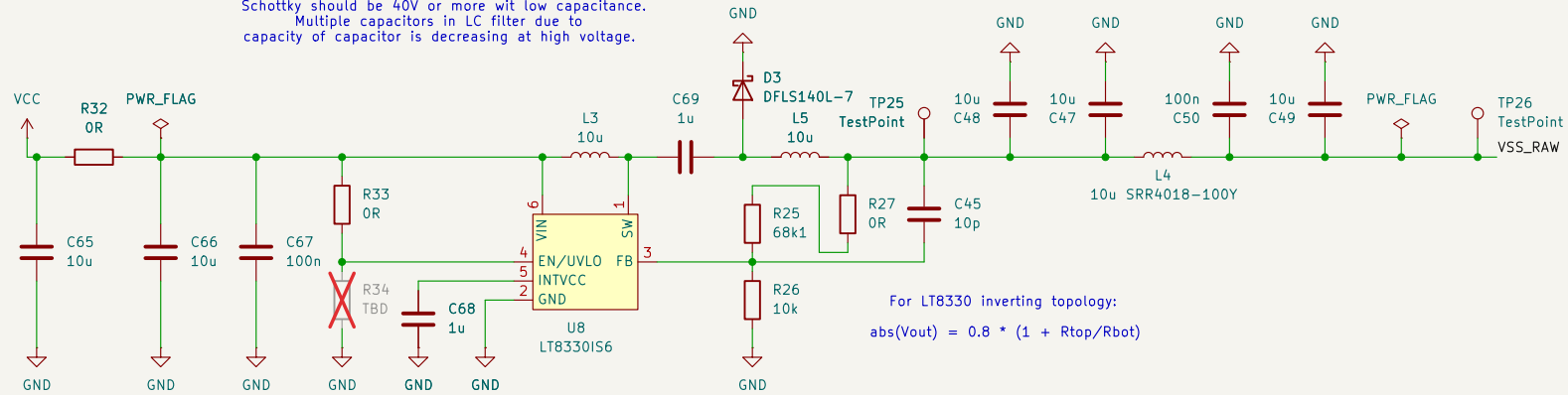
Rev:

Id: 6/13

1	2	3	4	5	6
A					A
B					B
C					C
D					D
1	2	3	4	5	6

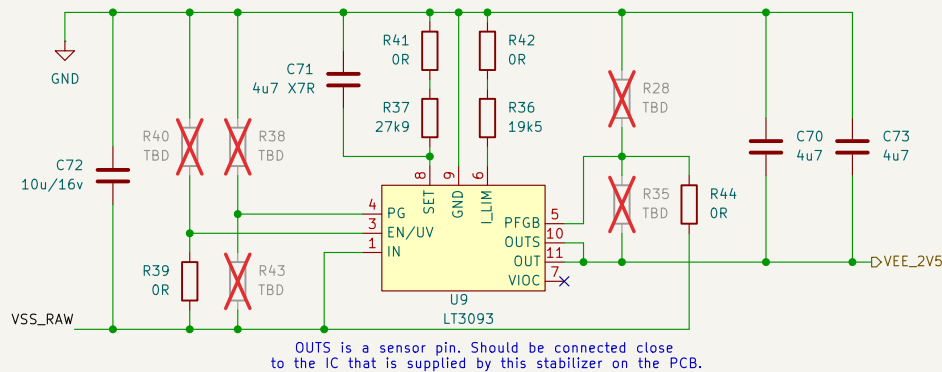
Sheet: /UserInterface/ File: UserInterface.kicad_sch		
Title:		
Size: A4	Date:	Rev:
KiCad E.D.A. 10.0.0		Id: 7/13

Schottky should be 40V or more wit low capacitance.
Multiple capacitors in LC filter due to
capacity of capacitor is decreasing at high voltage.



For LT8330 inverting topology:
 $abs(V_{out}) = 0.8 * (1 + R_{top}/R_{bot})$

Label	Direction	Description
VSS_RAW	Output	Raw negative voltage (-6.25V)
GND	Bidirectional	Ground



OUTS is a sensor pin. Should be connected close
to the IC that is supplied by this stabilizer on the PCB.

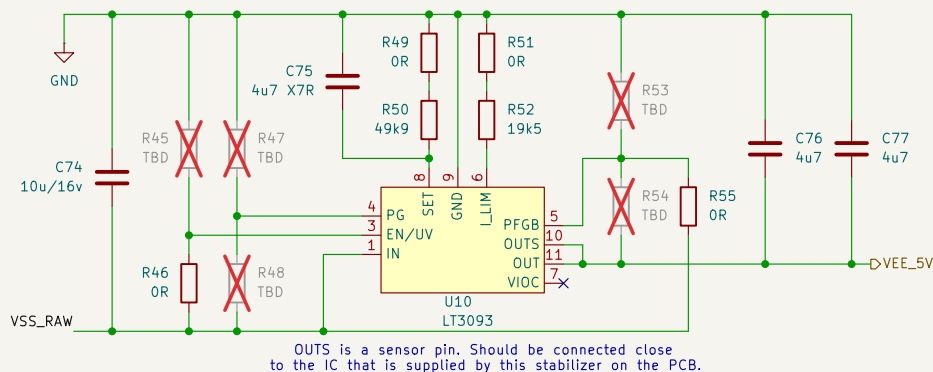
LT3093 calculation:

$$R_{set} = abs(V_{out}) / 100\mu A$$

$$I_{limit} = 1.95k\Omega \cdot A / R_{ilim}$$

for 70 mA (LTC6268): $R_{ilim} = 1.95k / 0.07 = 27.9k$
for 40 mA (AD8139): $R_{ilim} = 1.95k / 0.04 = 48.75k$

PG not used -> leave PG floating
PGFB not used -> tie PGFB to IN
EN/UV tied to IN for always-on operation



OUTS is a sensor pin. Should be connected close
to the IC that is supplied by this stabilizer on the PCB.

Sheet: /AnalogPowerNegative/
File: AnalogPowerNegative.kicad_sch

Title:

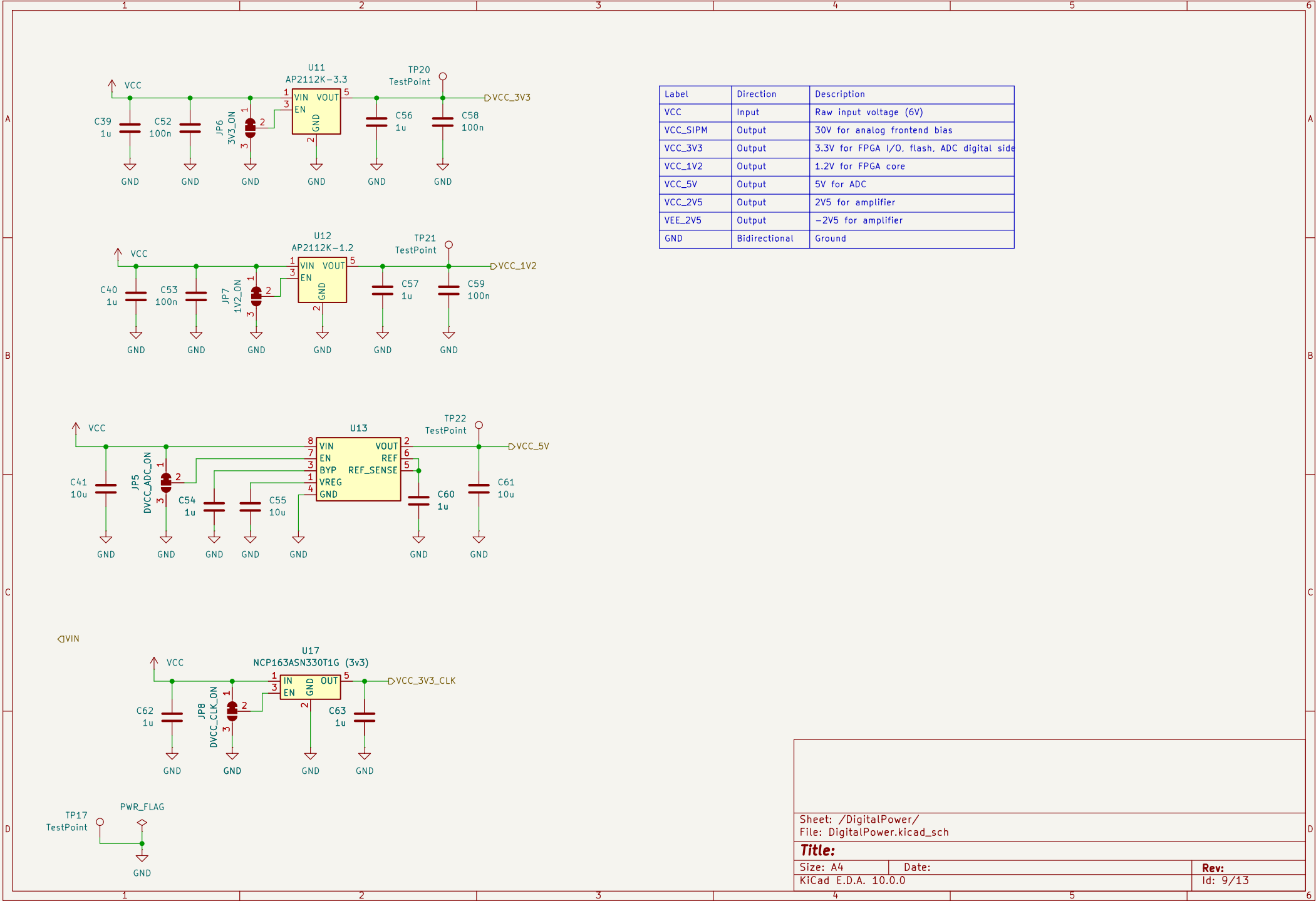
Size: A4

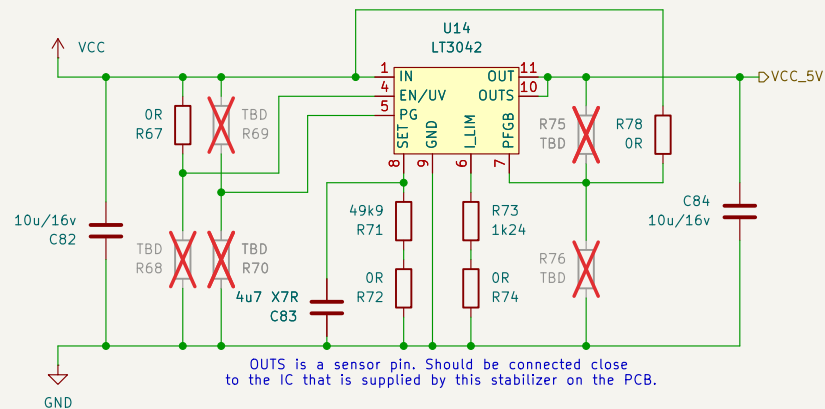
Date:

KiCad E.D.A. 10.0.0

Rev:

Id: 8/13





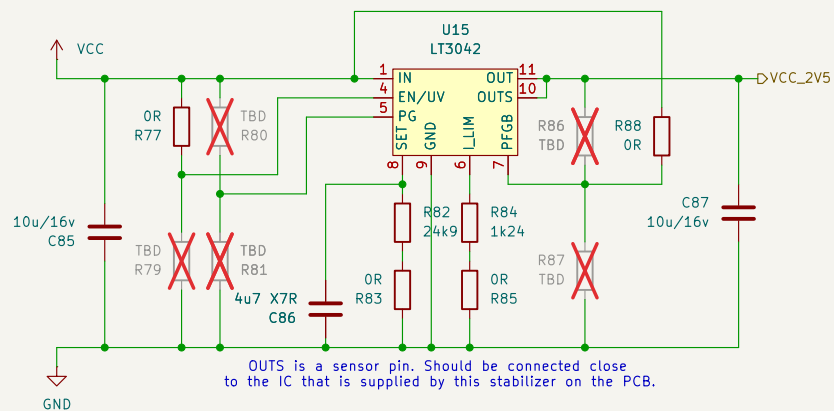
LT3042 calculation:

$$R_{set} = \frac{\text{abs}(V_{out})}{100\mu A}$$

$$I_{limit} = \frac{125\text{mA} \cdot k\Omega}{R_{ilim}}$$

For:
 for 60 mA: $R_{ilim} = \frac{125}{60} = 2.08k$
 for 70 mA: $R_{ilim} = \frac{125}{70} = 1.79k$
 for 100 mA: $R_{ilim} = \frac{125}{100} = 1.24k$

PG not used -> leave PG floating
 PGFB not used -> tie PGFB to IN
 EN/UV tied to IN for always-on operation



Sheet: /AnalogPowerPositive/
 File: AnalogPowerPositive.kicad_sch

Title:

Size: A4

Date:

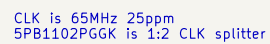
KiCad E.D.A. 10.0.0

Rev:

Id: 10/13

1	2	3	4	5	6
A					A
B					B
C					C
D					D
1	2	3	4	5	6

Sheet: /FPGAConfiguration/ File: FPGAConfiguration.kicad_sch		
Title:		
Size: A4	Date:	Rev:
KiCad E.D.A. 10.0.0		Id: 12/13



Sheet: /ClockGeneration/		D
File: ClockGeneration.kicad_sch		
Title:		
Size: A4	Date:	Rev:
KiCad E.D.A. 10.0.0		Id: 13/13